Invasive Ventilatory Support in Patients With Pediatric Acute Respiratory Distress Syndrome: From the Second Pediatric Acute Lung Injury Consensus Conference

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OBJECTIVE: To provide evidence for the Second Pediatric Acute Lung Injury Consensus Conference updated recommendations and consensus statements for clinical practice and future research on invasive mechanical ventilation support of patients with pediatric acute respiratory distress syndrome (PARDS).

DATA SOURCES: MEDLINE (Ovid), Embase (Elsevier), and CINAHL Complete (EBSCOhost).

STUDY SELECTION: We included clinical studies of critically ill patients undergoing invasive mechanical ventilation for PARDS, January 2013 to April 2022. In addition, meta-analyses and systematic reviews focused on the adult acute respiratory distress syndrome population were included to explore new relevant concepts (e.g., mechanical power, driving pressure, etc.) still underrepresented in the contemporary pediatric literature.

DATA EXTRACTION: Title/abstract review, full text review, and data extraction using a standardized data collection form.

DATA SYNTHESIS: The Grading of Recommendations Assessment, Development and Evaluation approach was used to identify and summarize relevant evidence and develop recommendations, good practice statements and research statements. We identified 26 pediatric studies for inclusion and 36 meta-analyses or systematic reviews in adults. We generated 12 recommendations, two research statements, and five good practice statements related to modes of ventilation, tidal volume, ventilation pressures, lung-protective ventilation bundles, driving pressure, mechanical power, recruitment maneuvers, prone positioning, and high-frequency ventilation. Only one recommendation, related to use of positive end-expiratory pressure, is classified as strong, with moderate certainty of evidence.

CONCLUSIONS: Limited pediatric data exist to make definitive recommendations for the management of invasive mechanical ventilation for patients with PARDS. Ongoing research is needed to better understand how to guide best practices and improve outcomes for patients with PARDS requiring invasive mechanical ventilation.

KEY WORDS: gas exchange; hypoxemia; mechanical ventilation; oxygenation; pediatric acute respiratory distress syndrome; ventilator induced lung injury

Mechanical ventilatory assistance is life-saving support for patients with pediatric acute respiratory distress syndrome (PARDS). However, its application is not without complications, and differences in the application of ventilatory strategies could influence disease outcomes. There have been numerous pediatric studies published since the 2015 Pediatric Acute Lung Injury Consensus Conference (PALICC) addressing the relationship between ventilator

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management and outcomes in children with PARDS (1). Furthermore, newer concepts related to mechanical ventilation, such as driving pressure (DP) and mechanical power (MP), have emerged from the adult ARDS literature.

In this article, we address key question no. 3 as outlined in the accompanying Methods article (2): "What is the effectiveness and comparative effectiveness of different ventilation strategies for children with PARDS?" We report the rationale and synthesis of evidence for recommendations related to invasive mechanical ventilation from the Second PALICC (PALICC-2).

METHODS

The details of the literature search are outlined in the PALICC-2 Methodology article in this supplement (2). A systematic review was conducted to identify relevant studies related to the management of mechanical ventilation in PARDS. In addition, since we also aimed to explore new relevant concepts, including MP and DP, that remain underrepresented in contemporary pediatric literature, we reviewed meta-analyses and systematic reviews conducted in the adult acute respiratory distress syndrome (ARDS) population. The complete search strategies can be found in the **Supplemental Tables 1** and **2** (http://links.lww.com/PCC/C304). Details of title/abstract review, full text review, and data extraction and generation of clinical practice recommendations, research statements, and policy statements are outlined in the PALICC-2 Methodology article (2).

RESULTS

Nonduplicate abstracts (2,720 pediatric and 1,590 adult) were identified and screened by the authors. Of these, 316 pediatric and 174 adult articles were identified for full text review (**Supplemental Fig. 1**, http://links.lww.com/PCC/C304). Based on the full text review, 26 pediatric and 36 adult articles were selected for full text extractions and were discussed by the authors to formulate and support a total of 19 clinical practice recommendations, and good practice statements and research statements. Supportive evidence-to-decisions are provided in the **Supplemental Tables 3–16** (http://links.lww.com/PCC/C304).

Modes of Ventilation

It remains unclear whether there is a relationship between ventilator mode and outcome in children with PARDS.

> **Research Statement 3.1.** We cannot make a recommendation on the specific ventilator mode preferred for PARDS patients. Future clinical studies should be conducted to assess controlled and assisted modes of ventilation on outcomes.
> (Ungraded research statement, 94% agreement)
> **Remarks:** There are no outcome data on the influence of mode (control or assisted, including airway pressure release ventilation [APRV], and neurally adjusted ventilatory assist [NAVA]) during conventional mechanical ventilation.

Justification

Two meta-analyses have been performed in adults with ARDS. Chacko et al (3) performed a meta-analysis of three randomized controlled trials (RCTs, n = 1089) comparing pressure control (PC) with volume control (VC) ventilation and found no significant difference in hospital mortality (risk ratio [RR] [95% CI] 0.83 [0.67–1.02]). There was a reduction in ICU mortality with PC ventilation (RR 0.84 [0.71–0.99]), but there is concern for residual confounding. The other is a meta-analysis of four RCTs (of acute hypoxemic respiratory failure, although not exclusively acute lung injury/ARDS patients) (4) that addressed the problem of confounding and showed no difference in ICU mortality based on mode of ventilation (odds ratio [OR] [95% CI] 1.01 [0.4–2.51]). One pediatric meta-analysis focused on different modes of ventilation, although the analyses primarily compared conventional modes of ventilation to high-frequency oscillatory ventilation (HFOV) (5). The mortality rate did not differ between the modes (OR 0.83 [95% CI 0.30–1.91]).

There has been one RCT in PARDS (6) comparing airway pressure release ventilation (APRV) with synchronized intermittent mandatory ventilation VC (low tidal volume [V_T]) with 26 subjects in each arm, 19% of them being classified as severe. The study was halted due to possible harm in the APRV group (RR of 28-d mortality 3.2 [95% CI 1–10.1]; $p = 0.081$). With the absence of studies showing superiority or even equivalence of APRV compared with low V_T ventilation with more traditional ventilatory modes, APRV cannot be recommended at this time. Overall, it remains unclear whether nontraditional ventilation modes, including APRV and neutrally adjusted ventilatory assist, have a role in the management of PARDS.

Tidal Volume

> **Recommendation 3.2.** We suggest the use of physiologic tidal volumes of 6-8 mL/kg in patients with PARDS as compared to supraphysiologic tidal volumes (V_T > 8 mL/kg). (Conditional recommendation, very low certainty of evidence, 98% agreement).
> Remarks: Tidal volume below 6 mL/kg should be used in patients, if needed, to remain below suggested plateau pressure (P_{Plat}) and driving pressure limits. Tidal volume below 4 mL/kg should be used with caution.

Justification

Two observational pediatric studies analyzed the effect of V_T limitation as part of a lung-protective ventilation bundle. A retrospective cohort based on the American-European consensus conference definition with a pre-post intervention design (7) showed higher mortality ($p = 0.04$) and fewer ventilator-free days (VFDs) ($p = 0.03$) for the before-implementation group compared with the after-implementation group. In the multivariable logistic regression analysis, higher V_T was independently associated with increased mortality (OR 1.59 [95% CI 1.2–2.1] for every mL/kg of V_T increase) and decreased VFD ($p = 0.03$). A second study (8) evaluated adherence to lung-protective ventilation guidelines based on the PALICC recommendations (1). Nonadherence to PALICC V_T recommendations was independently associated with higher mortality and longer duration of ventilation.

A meta-analysis of observational studies in children showed no relationship between V_T and mortality, regardless of V_T thresholds and presence/absence of PARDS (9). However, there are many limitations to this meta-analysis, including concerns about internal validity, high risk of bias, combining multivariable adjusted and crude (nonadjusted) results, and significant heterogeneity in results. Of note, there are several meta-analyses of interventional trials in adults with ARDS that proved a different conclusion. Petrucci et al (10) reported that ventilation with low V_T reduces 28-day mortality with a number-needed-to-treat of 10 (95% CI 6–25). Similarly, Parhar et al (11) performed a meta-analysis of 14 RCTs and quasi-RCTs with multivariable meta-regression and found a relationship between V_T reduction and reduction in mortality ($p = 0.043$). However, additional analyses highlight the importance of other ventilator variables in the relationship between V_T and outcome. Amato et al (12) performed meta-regression and identified that the combination of higher V_T (median of 8 mL/kg) with higher DP groups (> 15 cm H_2O) is associated with higher mortality. Walkey et al (13) pooled data from RCTs with V_T reduction versus control strategies and revealed that V_T reduction is effective only within a combined open lung approach.

Furthermore, there may be risk when V_T is ultra-low. Eichacker et al (14) demonstrated a "U-shape" curve in the association between V_T and mortality, with higher mortality at both low (typically < 4 mL/kg) and high V_T. Costa et al (15) demonstrated the same "U-shape" curve, further highlighting that with lower compliance, the optimal range of V_T size shifted to the left (i.e., smaller V_T).

Benefits and Risks

On balance, a "physiologic range" of V_T can result in lower mortality by preventing ventilator-induced lung injury. Potential risks for limiting the upper range of V_T to 8 mL/kg include potential patient-ventilator asynchrony, particularly for patients with high respiratory drive, and inadequate gas exchange. There also appears to be higher mortality from ultra-low V_T (perhaps < 4 mL/kg), and the most appropriate lower limit is unknown in children.

Implementation Considerations

The implementation of physiologic V_T between 6 and 8 mL/kg is feasible in children with PARDS and may improve outcomes. Further reduction in V_T below 6 mL/kg may be needed to maintain lung-protective limits on DP (see next section) and plateau pressure (P_{Plat}). The balance of effects favors the use of this physiologic range of V_T, although some concern is raised by the possibility of a U-shape curve in the relationship between V_T and mortality, and the precision of the limits in children.

Ventilation Pressures

Recommendation 3.3.1. In the absence of transpulmonary pressure measurements, we suggest an inspiratory $P_{Plat} \leq 28$ cm H_2O.
(Conditional recommendation, very low certainty of evidence, 92% agreement).
Remarks: The P_{Plat} may be higher (29-32 cm H_2O) for patients with reduced chest wall compliance.

Justification

Neither RCTs nor observational studies have been performed in children with PARDS evaluating the relationship between P_{Plat} (16) and outcome. In the adult literature, Petrucci et al (10) found in a stratified analysis that the reduction in mortality with low V_T was observed only in the cohort of studies with an experimental group ventilated with P_{Plat} greater than 31 cm H_2O, whereas there was no effect if the control group had a P_{Plat} less than 31 cm H_2O.

Benefits and Risks

Limiting P_{Plat} may have a notable benefit in reducing mortality. Potential risks of limiting P_{Plat} include inadequate gas exchange, potential higher use of sedation or neuromuscular blockade to maintain P_{Plat} limits, or underutilization of positive end-expiratory pressure (PEEP) if it is reduced to maintain P_{Plat} limits.

Implementation Considerations

P_{Plat} should be measured during an inspiratory pause on the ventilator (see accompanying article in this supplement about monitoring [16]), a practice which may not be common in many PICUs. Clinicians must be trained to interpret the pressure waveform to ensure a plateau has been reached. Although the peak pressure is often measured, the relationship between peak and P_{Plat} is confounded by mode of ventilation, flow characteristics, airway resistance, and patient effort. Peak pressure can therefore under- or overestimate P_{Plat}. It remains unclear if limiting DP (P_{plat}–PEEP) improves clinical outcomes for children with PARDS.

Recommendation 3.3.2. We suggest limiting DP to 15 cm H_2O (as measured under static conditions) for patients with PARDS.
(Conditional recommendation, very low certainty of evidence, 82% agreement).

Justification

There are no RCTs in children, although several observational studies have been published. In a retrospective cohort of 380 children, Rauf et al (17) found DP greater than 15 cm H_2O was independently associated with morbidity (defined as presence of both duration of ventilation > 7 d and PICU length of stay > 10 d or death) after adjusting for confounding variables (adjusted OR for morbidity 3.2 [95% CI 1.2–9]; $p = 0.02$). The optimal cutoff was 19 cm H_2O (area under the receiver operating characteristic curve 0.8). However, there was no difference in mortality between the two groups (17% in low DP vs 24% in high DP; $p = 0.38$).

Díaz et al (18) described pulmonary mechanics in 30 children under general anesthesia and 38 children with PARDS. DP was higher in the PARDS group compared with the anesthesia group, and DP was the variable that best discriminated PARDS from non-PARDS in children studied. van Schelven et al (19) demonstrated in patients ventilated for a direct pulmonary indication that higher DP was independently associated with longer time to extubation ($p < 0.001$) after adjusting for Pediatric Risk of Mortality-III 24-hour score and oxygenation index. Higher DP was also associated with fewer 28-day VFDs.

In adults, Amato et al (12) pooled data from RCTs with multivariable adjustment for confounding (multilevel mediation analysis) and reported that among ventilation variables, DP was most strongly associated with survival. A 1 SD increase in DP (approximately 7 cm H_2O) was associated with increased mortality (RR 1.41 [95% CI 1.31–1.51]; $p < 0.001$), even in patients receiving "protective" P_{Plat} and V_T. Individual changes in V_T or PEEP after randomization were not independently associated with survival, except if they were among changes leading to reductions in DP (mediation effects of DP, $p = 0.004$ and $p = 0.001$, respectively). Other meta-analyses (20, 21) in the adult ARDS population have demonstrated similar findings.

Benefits and Risks

The desirable effects of the use of a limited DP strategy may be considered large given the potential impact on mortality. However, although it is clear that patients with higher DP have worse outcomes, the best strategy to manage these patients is not established. Similar to P_{Plat} limitation, undesirable effects of lowering DP include potential for more use of sedation or neuromuscular blockade, inadequate gas exchange, and potentially higher use of adjuvant therapies to stay below the DP limits.

Implementation and Other Considerations

DP (P_{Plat}–PEEP) should be measured under static conditions. This includes an inspiratory pause to measure P_{Plat} and an expiratory pause to ensure there is no intrinsic PEEP. Although the data from both adult and pediatric studies suggest a DP limit of 13–15 cm H_2O, this should be evaluated in future clinical trials.

PEEP

Recommendation 3.4.1 We suggest that PEEP should be titrated to oxygenation/oxygen delivery, hemodynamics, and compliance measured under static conditions as compared to other clinical parameters or any of these parameters in isolation in patients with PARDS.
(Conditional recommendation, very low certainty of evidence, 96% agreement).

Recommendation 3.4.2 We recommend that PEEP levels should typically be maintained at or above the lower PEEP/higher F_{IO_2} table from the ARDS Network protocol (22) (**Supplemental Table 17**, http://links.lww.com/PCC/C304) as compared to PEEP levels lower than the lower PEEP/ higher F_{IO_2} table.
(Strong recommendation, moderate certainty of evidence, 96% agreement).

Good Practice Statement 3.4.3. When adjusting PEEP levels to achieve the proposed oxygen target range for PARDS, attention to avoid exceeding plateau pressure and/or DP limits is warranted.
(Ungraded good practice statement, 100% agreement)

Justification

Observational studies report an association of higher mortality when applied PEEP is lower than the ARDS Network (ARDSNet) low PEEP/high F_{IO_2} table as compared to on-protocol or PEEP higher than protocol (Khemani et al [23] OR 2.05 [95% CI 1.32–3.17]; Bhalla et al [8] OR 1.12 [95% CI 1.01–1.23]). Bhalla et al (8) also found that on-protocol or PEEP higher than protocol was associated with shorter duration of mechanical ventilation ($p = 0.01$). Wong et al (7) reported the beneficial effects of implementing a lung-protective ventilation protocol which resulted in higher PEEP, mainly in the moderate-severe PARDS category (10 [95% CI 8–12] vs 9 cm H_2O [95% CI 8–10 cm H_2O]; $p = 0.011$) and a lower V_T. The protocol was associated with a significant reduction in mortality (adjusted hazard ratio [HR], 0.37 [95% CI 0.16–0.88]; $p = 0.025$) as compared to the preimplementation period.

Although a PEEP strategy based on oxygenation (such as the ARDSNet PEEP/F_{IO_2} table [22]) is reasonable, adult data highlight that the benefits of PEEP titration should also evaluate improvements in other variables, namely compliance (or DP) and hemodynamics. Yehya et al (24) reported that after manipulations to PEEP, changes in DP were more strongly associated with mortality than changes in Pa_{O_2}/F_{IO_2}, which highlights that a measure of lung recruitment (improvement in compliance or reduction in DP with same V_T) is important to evaluate changes in PEEP.

Benefits and Risks

There is a moderate certainty of evidence regarding the application of the ARDS network PEEP/F_{IO_2} grid to those with PARDS. The available studies have a low risk of bias and no inconsistency or imprecision in results. However, a multifaceted approach should be used to identify the optimal PEEP level for an individual patient with PARDS. PEEP is often applied to improve oxygenation (i.e., by lung volume recruitment), and intensivists tend to use only oxygenation to assess the accuracy of PEEP titration. However, increasing PEEP may also lead to overdistension of nonrecruited lung regions. This may lead to improvements in oxygenation, but a reduction in oxygen delivery or a compromise of lung mechanics. Hence, PEEP should be titrated to balance respiratory system compliance, hemodynamic status, and oxygen delivery. Undesirable effects associated with the use of higher levels of PEEP include the potential for worsening ventilator-induced lung injury if lung units are overdistended, particularly if P_{Plat} or DP limits are exceeded. Hemodynamic instability is also possible, and hemodynamics must be monitored closely during PEEP titration.

Implementation Considerations

Implementing a protocol to set PEEP at/or higher than the ARDS Network PEEP/FIO_2 grid should be feasible in most PICUs. However, there is high variability in further methods for PEEP titration, with no consensus nor evidence for the best approach. Local protocols should be developed to have a consistent framework for PEEP selection which evaluates changes in respiratory system compliance, oxygen delivery, and hemodynamics.

Ventilation Bundle

In patients with PARDS, the use of a lung-protective ventilation bundle versus nonadherence to such a bundle may improve critical outcomes.

> **Recommendation 3.5.** We suggest that a lung protective ventilation bundle should be used as compared to no bundle when caring for ventilated patients with PARDS.
> (Conditional recommendation, low certainty of evidence, 83% agreement).
> Remarks: Lung protective bundles prioritize maintaining multiple ventilator settings (V_T, P_{Plat}, DP, PEEP) within the suggested limits as well as using detection algorithms and educational programs.

Justification

Only observational studies in children are available. Wong et al (7) performed a prospective before-and-after comparison study of a lung-protective mechanical ventilation bundle protocol (low FIO_2, low V_T, PEEP/FIO_2 grid). These investigators observed a decrease in mortality, adjusted HR 0.37 (95% CI 0.16–0.88) (p = 0.025), with no increase in VFD and PICU-free days after protocol implementation. Bhalla et al (8) analyzed a prospective cohort of children with PARDS and identified that higher nonadherence to V_T and PEEP recommendations were independently associated with higher mortality and longer duration of mechanical ventilation in a multivariable logistic regression and competing-risks regression.

In the adult ARDS population, Parhar et al (11) pooled data from 14 RCTs and quasi-RCTs in a meta-analysis showing a nonsignificant reduction (with high heterogeneity) in mortality (RR 0.83 [95% CI, 0.52–1.33]; $p = 0.439$; $I^2 = 67\%$) with lung-protective bundles.

Benefits and Risks

The desirable effects of a lung-protective ventilation bundle approach are decreased mortality and shorter duration of ventilation. There are likely minimal undesirable effects from implementation of a standardized bundle to maintain within recommended lung-protective limits, although a bundled protocol should not replace clinical judgment to individualize selection of ventilator settings.

Implementation Considerations

The composite evidence favors the use of lung-protective ventilation bundles. The intervention could lower mortality and/or ventilation days. The implementation of such bundles should consider the local environment and may benefit from the use of electronic tools (see the accompanying article on Clinical Informatics and Data Science [25]).

Recruitment Maneuvers

> Recommendation 3.6. We cannot suggest for or against the use of recruitment maneuvers in patients with PARDS.
> (Conditional Recommendation, low certainty of evidence, 94% agreement).
> Remarks: Careful recruitment maneuvers may be applied in the attempt to improve oxygenation by slow incremental and decremental PEEP steps. Sustained inflation maneuvers cannot be recommended.

Justification

Two RCTs in adults with ARDS have identified (26, 27) safety concerns with the use of stepwise recruitment maneuvers (RMs) (i.e., incremental PEEP increases of

5 cm H_2O up to 25–30 cm H_2O), followed by decremental PEEP titration. These studies demonstrated higher rates of barotrauma and hemodynamic compromise, with higher mortality. Meta-analyses combining adult RCTs studying RM in adults with ARDS (28–30) show no consistent difference in mortality; however, RMs did result in reduction in the use of rescue therapies for hypoxemia (RR 0.69 [95% CI 0.56–0.84]) (29). RMs were associated with higher rates of hemodynamic compromise.

There are no RCTs studying RMs in PARDS. Three clinical studies in 57 pediatric patients with ARDS reported safe use of a staircase RM approach (Boriosi et al [31], Cruces et al [32], Kheir et al [33]). A sustained 4–24-hour improvement in oxygenation and a reduction in $Paco_2$ was reported in two of these studies (31, 32), but not the other (33). Effects on dynamic compliance and functional residual capacity were variable.

Benefits and Risks

RMs can prevent lung collapse and improve oxygenation and attenuate derecruitment-associated lung injury. The pediatric studies demonstrated that these children tolerated RM without hemodynamic compromise (i.e., hypotension and/or bradycardia), with no significant barotrauma, hypoxemia, or arrhythmia, although transient hypercapnia and respiratory acidosis have been reported. However, the data from adult RCTs highlight there may be harm introduced if aggressive RMs are used, particularly in patients who do not have recruitable lung.

Implementation Considerations

The above recommendations and comments are based on a low certainty of evidence. The pediatric studies were generally designed only to evaluate impact on gas exchange, pulmonary mechanics, and safety but not mortality, PICU length of stay, or duration of mechanical ventilation. Given the potential risks noted in adult studies when large PEEP steps were used (5 cm H_2O), we suggest that in selected patients with severe PARDS, slow incremental/decremental changes of PEEP (typically PEEP steps of 2 cm H_2O) may be performed while continually monitoring oxygenation, compliance, and hemodynamics. Sustained inflation maneuvers (i.e., elevation of the pressure above the point of hyperinflation and maintained for an extended period, generally 30–40 s in adult patients) cannot be recommended for PARDS.

Mechanical Power

Research Statement 3.7. There is insufficient evidence to suggest the use of mechanical power calculations to guide pediatric mechanical ventilation. Future research is needed with regard to mechanical power calculations in children before clinical use.
(Ungraded research statement, 90% agreement).

Justification

Only a few observational studies have evaluated MP in PARDS. Proulx et al (34) found that MP was associated with mortality after adjustment for respiratory rate, age, and daily Pediatric Logistic Organ Dysfunction (PELOD)-2 score (mean difference in MP between nonsurvivors and survivors = +3 Joules/breath [1–6 Joules/breath]; $p = 0.009$). Bhalla et al (35) found an association between higher MP and fewer VFD, after controlling for confounding variables in a multivariable competing risk analysis (per 0.1 $J \cdot min^{-1} \cdot kg^{-1}$ subdistribution HR [SHR] 0.93 [0.87–0.98]; $p = 0.013$). Higher MP was associated with higher mortality when children who died for neurologic reasons were excluded (per 0.1 $J \cdot min^{-1} \cdot kg^{-1}$ OR = 1.22 [95% CI = 1.01–1.46], $p = 0.036$). In subgroup analyses by age, the association between higher MP and fewer VFD remained only in children less than 2 years old (per 0.1 $J \cdot min^{-1} \cdot kg^{-1}$ SHR = 0.89 [95% CI = 0.82–0.96]; $p = 0.005$). Of note, because VC ventilation is less frequently used in pediatrics, modifications to the original MP formulas are typically used in pediatric studies (36, 37).

In adults with ARDS, Tonna et al (38) carried out a meta-analysis using individual data from RCTs and showed that MP retains a significant relationship with mortality despite adjusting for DP, which was possibly due to the reliance of MP on components other than DP itself. Thus, MP captures an applied energy in a way that DP does not. Zhang et al (39) pooled data from eight RCTs and found an interaction in the effect on mortality of MP (normalized to predicted body weight) and ARDS severity. Modesto i Alapont et al (40) analyzed 14 RCTs based on an open lung strategy. Pooled analysis with meta-regression highlighted that open lung strategies had a beneficial impact on mortality when MP or DP was lower in the intervention group, as compared to the control group. Pediatric specific research is needed to identify if MP should be a therapeutic target for ventilator management in PARDS.

High-Frequency Ventilation

Recommendation 3.8.1. We cannot make a recommendation as to whether high frequency oscillatory ventilation (HFOV) should be used instead of conventional ventilation in patients with PARDS.
(Conditional recommendation, low certainty of evidence, 90% agreement).
Remarks. HFOV may be considered in patients with PARDS in whom the ventilatory goals are not achieved with a lung protective strategy on conventional mechanical ventilation.
Good Practice Statement 3.8.2. When using HFOV, the optimal lung volume should be achieved by exploration of the potential for lung recruitment with a stepwise increase and decrease of the mean airway pressure under continuous monitoring of the oxygen and CO_2 response and hemodynamic parameters.
(Ungraded good practice statement, 90% agreement).

Justification

Pediatric evidence to support this recommendation is limited, as the results of three RCTs are inconclusive with no reported difference in mortality (41–43). El-Nawawy et al (43) reported no difference in VFD, whereas Arnold et al (41) reported a significant difference in oxygen requirement at 30 days (13.7% vs 34%; $p = 0.039$). It must be noted that the study by Arnold et al (41) was performed prior to the routine use of low V_T ventilation (**Fig. 1**).

Observational studies with propensity score analysis of matched patients managed with HFOV compared with conventional mechanical ventilation are variable (44–46). Bateman et al (44) reported no difference in mortality (OR 1.28 [95% CI 0.92–1.79]; $p = 0.15$) but did show longer duration of mechanical ventilation with early use of HFOV. Gupta et al (45) demonstrated a longer duration of mechanical ventilation, longer ICU length of stay, and higher mortality in HFOV and early HFOV patients compared with those conventionally ventilated. Wong et al (46) similarly showed an association of increased mortality with HFOV but no difference in VFD (**Fig. 2**).

Experimental studies show the benefit of lung volume optimization at HFOV initiation, which was not done in previous HFOV trials. Two observational studies demonstrate the benefit of a lung volume optimization strategy (47, 48), with an improvement in oxygenation, pH, and $Pa{CO_2}$ (47). The pulse oximetry saturation to $F{IO_2}$ ratio ($Sp{O_2}/F{IO_2}$) had a sensitivity of 93% and a specificity of 77% to detect recruitment 1 hour after the HFOV lung volume optimization maneuver (48). The currently enrolling Prone and Oscillation Pediatric Clinical Trial (NCT03896763) may alter this recommendation over time.

Benefits and Risks

Currently available data show no clear improvement in clinical outcome with the use of HFOV in the management of PARDS. Clinical studies have reported an increased need for neuromuscular blockade (41–43, 47, 48). Retrospective analyses suggest some association of increased mortality and duration of mechanical ventilation with HFOV. However, the certainty of evidence is low, and concern for residual confounding remains.

Certainty of Evidence

The overall certainty of evidence of this recommendation is low as there was serious risk of bias in both the RCTs and the observational studies (**Table 1**) (Supplemental Table 12, http://links.lww.com/PCC/C304). Notably, none of the RCTs applied the PARDS PALICC definition. Arnold et al (41) did not employ a lung-protective strategy. Samransamruajkit et al (42) has a risk of bias due to cointervention with a RM as well as a high cross-over rate. El-Nawawy et al (43) reported very high mortality in both groups, which raises concerns about generalizability. Although the observational studies conducted propensity score analyses, there is concern for residual confounding.

Implementation Considerations

The lack of data to support routine use of HFOV in PARDS precludes a recommendation as to whether HFOV should be used instead of conventional ventilation in the management of patients with PARDS. However, it may be considered in circumstances where adequate gas exchange cannot be maintained with conventional lung-protective ventilation. If used, we suggest a procedure for lung volume optimization with initiation of HFOV. The optimal methods for adjusting HFOV settings and the identification of the patients who may best benefit remain an important research question, and a large multinational clinical trial is ongoing. HFOV also requires the associated expense of an additional ventilator and staff training.

Oxygenation Target

Recommendation 3.9.1. We suggest that for mild/moderate PARDS, Spo_2 be maintained between 92–97%.
(Conditional recommendation, low certainty of evidence, 98% agreement).

Recommendation 3.9.2. We suggest that for severe PARDS, after optimizing PEEP, Spo_2 less than 92% can be accepted to reduce Fio_2 exposure.
(Conditional recommendation, low certainty of evidence, 88% agreement).

Good Practice Statement 3.9.3. Prolonged exposure to hypoxemic (<88%) or "high" (>97%) Spo_2 targets should be avoided while on oxygen supplementation.
(Ungraded good practice statement, 88% agreement).

Good Practice Statement 3.9.4. When Spo_2 is < 92%, central venous saturation and markers of oxygen delivery/utilization should be monitored.
(Ungraded good practice statement, 94% agreement).

Justification

In pediatric patients with ARDS, improvements in oxygenation do not always translate into improved clinical outcomes (49, 50). Furthermore, the implementation of lung-protective strategies is typically associated with lower Pao_2 and lower mortality in adult and pediatric populations (7, 51–53). Wong et al (7) reported the increased use of permissive hypoxemia (Spo_2 88–92%) in moderate-severe PARDS in a protocol versus non-protocol group (29% vs 17%; $p = 0.031$) with a reduction in median (interquartile range [IQR]) Pao_2 (64.7

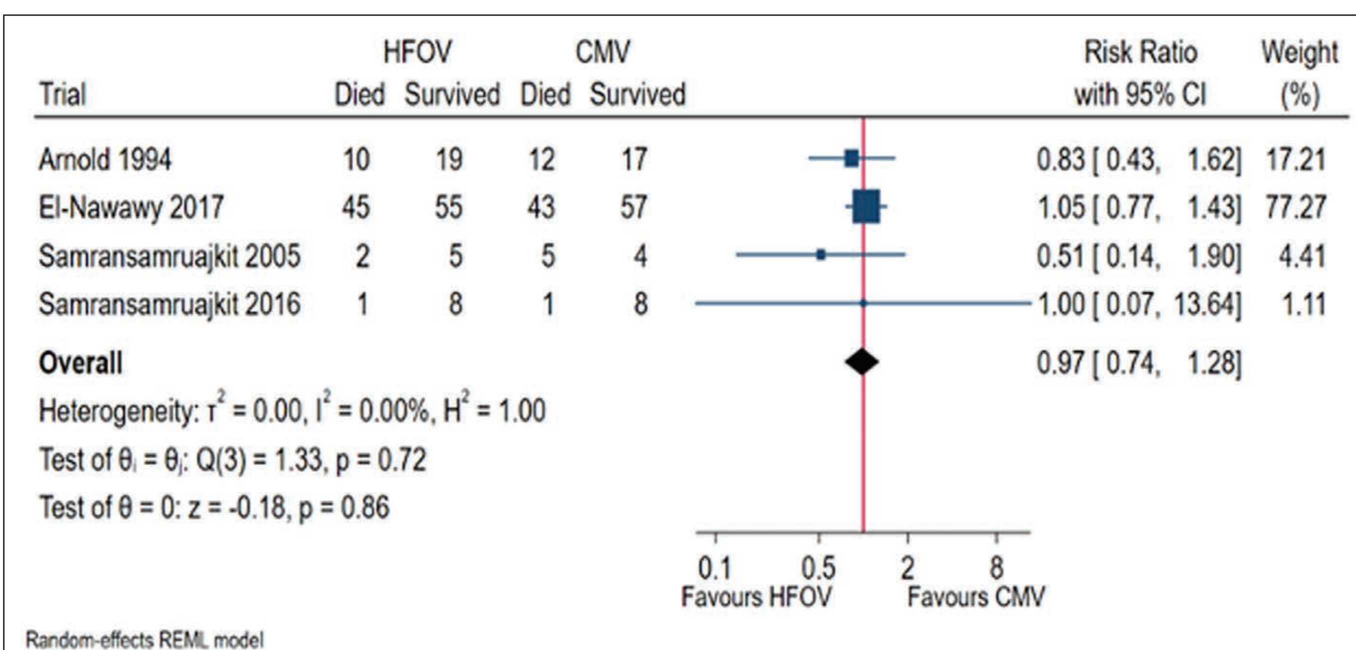

Figure 1. Forest plots comparing high-frequency oscillatory ventilation (HFOV) and conventional mechanical ventilation (CMV) in children with pediatric acute respiratory distress syndrome for the outcome of all-cause mortality in randomized control trials. REML = restricted maximum likelihood.

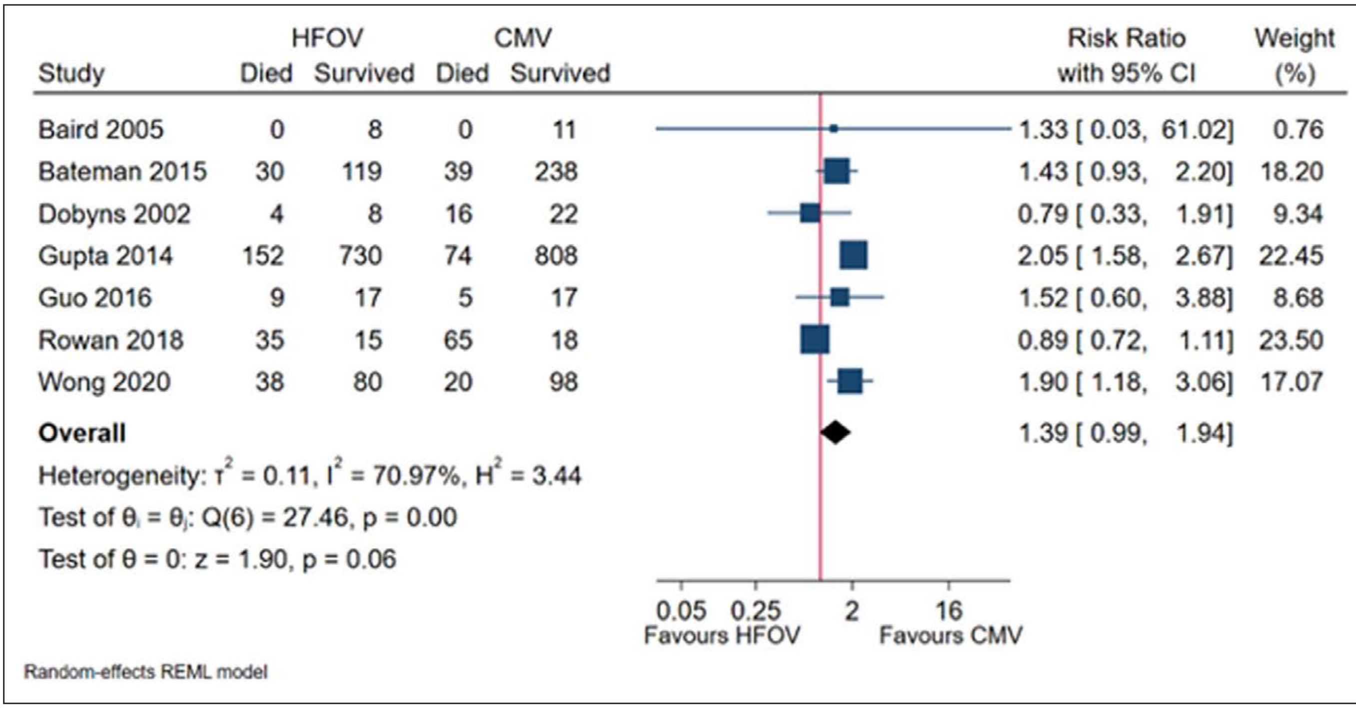

Figure 2. Forest plots comparing high-frequency oscillatory ventilation (HFOV) and conventional mechanical ventilation (CMV) in children with pediatric acute respiratory distress syndrome for the outcome of all-cause mortality in observational studies. REML = restricted maximum likelihood.

[57.8–80.5] vs 75.3 mm Hg [62.2–88.0 mm Hg]; p = 0.015) after protocol implementation. After adjusting for the Pediatric Index of Mortality–2 score, PELOD score, and PARDS severity in the multivariable Cox proportional hazard model, the lung-protective ventilation protocol was associated with a lower mortality risk (adjusted HR, 0.37 [95%CI 0.16–0.88]; p = 0.025). Because this is a bundled intervention, it is unclear how much direct benefit is coming from different Pa_{O_2}.

Observational data suggest that there may be a U-shaped relationship between mortality and oxygen. This suggests that both hypoxia and hyperoxia should be avoided when possible. Sp_{O_2} lower than 88% for prolonged periods is associated with higher mortality in critically ill children (54). Crude mortality was 15.4% with hypoxia, 5.3% with normoxia, and 9.1% in hyperoxic patients with a standardized mortality rate of 0.93, 0.75, and 0.82 for hypoxic, normoxic, and hyperoxic patients, respectively. Systematic review showed OR 3.13 (95% CI 1.79–5.48; p < 0.001) of mortality with hypoxia compared with normoxia.

Benefits and Risks

The benefits of maintaining Sp_{O_2} to simultaneously prevent both hyperoxia and hypoxia include limiting unnecessary oxygen exposure while preventing inadequate oxygen delivery to the tissues. This approach has the potential to improve clinical outcomes. Risks however include imprecision of defining the thresholds for both hypoxemia and hyperoxia based on pulse oximetry, with limited data about the long-term neurologic or other end-organ effects of permissive hypoxemia. As such, clinicians must weigh the potential benefits and risks of this approach for each individual clinical situation, and when permissive hypoxemia is being used, specific monitoring of central venous saturation to assess oxygen delivery is recommended.

Certainty of Evidence

There is a low certainty of evidence on the appropriate ranges of oxygenation in patients with severe PARDS. The concern regarding prolonged exposure to hypoxemia relies on indirect evidence.

Implementation Considerations

The implementation of the proposed oxygenation targets likely requires multidisciplinary approaches empowering bedside providers (i.e., nurses, respiratory therapists) to adjust the level of oxygen to maintain the therapeutic targets and adjust bedside alarms to reflect the target range. Automated systems such as closed-loop oxygen controllers may also improve compliance.

TABLE 1.
Pediatric Studies Comparing High-Frequency Oscillatory Ventilation Versus Conventional Mechanical Ventilation (Randomized Controlled Trials and Propensity Score Analysis)

Study Design	Patients (*n*)	Cross-Over HFO vs CMV, *n* (%)	Length of Mechanical Ventilation HFO vs CMV days	Neuro muscular Blockade HFO vs CMV, *n* (%)	Outcomes HFO vs CMV, *n* (%)	Mortality HFO vs CMV, *n* (%)
Arnold et al (41) RCT 1990/1994	$n = 58$ 29 HFO/29 CMV	11(38) vs 19 (66)	20 ± 27 vs 22 ± 17	–	Number of patients requiring O_2 at 30 d of observation 4 (13.7) vs 10 (34)	10 (34) vs 12 (41)
Samransamruajkit et al (42) RCT 2012/2013	$n = 18$ 9 HFO/9 CMV	6/9 CMV to HFO	–	–	–	1 (11) vs 1 (11)
El-Nawawy et al (43) RCT 2011/2016	$n = 200$ 100 HFO/100 CMV	0 (0) vs 16 (16.0) $p < 0.001$	25.0 (23.0–26.0) vs 25.0 (23.0–26.0) $p = 0.658$	85 (85) vs 10 (10) $p = 0.001$	–	45 (45.0) vs 43 (43.0) $p = 0.7$
Gupta et al (45) Propensity score 2009/2011 Database United States	All HFO 1,764 882 HFO/882 CMV Early HFO 941 early HFO 471/471 CMV		All HFO 20.3 (19.7) vs 14.6 (15.4) $p < 0.001$ Early HFO 15.9 (15.9) vs 14.6 (16.4) $p < 0.001$			All HFO 152 (17.3) vs 74 (8.4) $p < 0.001$ Early HFO 85 (18.1) vs 39 (8.3) $p < 0.001$
Bateman et al (44) Propensity score 2009/2013 (Randomized Evaluation of Sedation Titration for Respiratory Failure)	Quintile 5 $n =$ 213 99 early HFO/114 CMV+HFO		14.9 (9.7–28.0) vs 10.8 (5.8–24.9) $p = 0.08$	53 (33–75) vs 25 (0–56) $p = 0.001$		25 (25) vs 19 (17) $p = 0.09$
Wong et al (46) Propensity score 2009/2015 10 PICUs Asia	$n = 236$ 138 HFO/138 CMV		Ventilator-free days 4.0 (0.0–16.0) vs 4.0 (0.0-17.8) $p = 0.26$		PICU-free days 0.0 (0.0–11.0) vs 4.0 (0.0–15.8) $p = 0.03$	38 (32.2) vs 20 (16.9) $p = 0.01$

HFO = high-frequency oscillatory ventilation, RCT = randomized controlled trial, CMV = controlled mechanical ventilation.
Dashes indicate data not available.

pH Target

Recommendation 3.10.1. We suggest allowing permissive hypercapnia (to a lower limit pH of 7.20) in patients with PARDS to remain within previously recommended pressure and tidal volume ranges.

(Conditional recommendation, very low certainty of evidence, 100% agreement).

Remarks: Exceptions to permissive hypercapnia include, but are not limited to, intracranial hypertension, severe pulmonary hypertension, select congenital heart disease lesions, hemodynamic instability, and significant ventricular dysfunction.

Recommendation 3.10.2. We suggest against the routine use of bicarbonate supplementation as compared to selective use of bicarbonate in PARDS.

(Conditional recommendation, low certainty of evidence, 96% agreement).

Remarks: Bicarbonate supplementation can be considered in situations where severe metabolic acidosis or pulmonary hypertension are adversely affecting cardiac function or hemodynamic stability.

Justification

The decrease in minute ventilation associated with low V_T ventilation, coupled with increased dead space ventilation in patients with ARDS, may lead to respiratory acidosis. In the pediatric population, two recent studies support the use of permissive hypercapnia. Wong et al (7) implemented a lung-protective ventilation strategy as part of a quality improvement initiative and observed a significant reduction of V_T. The median (IQR) $Paco_2$ significantly increased (53.0 [45.2–63.5] vs 46.2 mm Hg [39.7–54.9 mm Hg]; $p < 0.001$) in association with the protocol; however, no significant difference in the median (IQR) pH (7.32 [7.24–7.39] vs 7.36 [7.28–7.41]; $p = 0.086$) was seen. The lung-protective ventilation protocol was associated with a lower mortality risk (adjusted HR, 0.37 [95% CI 0.16–0.88]; $p = 0.025$). In a retrospective analysis of two cohorts of immunocompromised patients with PARDS (pre- and postimplementation of a lung-protective protocol embracing permissive hypercapnia), Fuchs et al (55) observed a significant improvement in survival (59% vs 75%; $p < 0.005$) with significantly higher $Paco_2$ levels, even as high as 120–140 mm Hg in some patients. Of note, patients had appropriate metabolic compensation for the higher $Paco_2$ because median pH was 7.35 (7.26–7.46).

Benefits and Risks

Lung-protective ventilation is associated with lower mortality and inherently necessitates permissive hypercapnia to maintain lung-protective limits. Potential risks of permissive hypercapnia include higher use of neuromuscular blockade or higher use of sedation to blunt respiratory drive to tolerate the hypercapnia, which could have detrimental effects related to immobility, risk of infection, delirium, and long-term neurodevelopmental outcome.

Implementation Considerations

Practitioners should consider the risks and benefits for the level of hypercapnia which is permissible for a given patient. This consideration should include weighing the benefits of preventing ventilator-induced lung injury from limiting high ventilator settings against potential risks of higher levels of sedation and neuromuscular blockade. Patients with significant hemodynamic or neurologic impairment may have additional risks with permissive hypercapnia. There are also no clear lower limits of pH available in the literature to define "permissive hypercapnia," and pH levels above 7.2 should be generally well accepted amongst pediatric intensivists.

Endotracheal Tubes

Cuffed endotracheal tubes (ETTs) are generally beneficial to maintain mean airway pressure and avoid inadvertent derecruitment. Consideration should be given to the different types of cuffed ETT available (e.g., low profile cuffs).

Good practice statement 3.11. Cuffed endotracheal tubes should be used when ventilating a patient with PARDS.

(Ungraded good practice statement, 100% agreement).

Justification

Modern cuffed ETTs have not been associated with higher rates of complications than uncuffed ETTs in children regardless of age, as long as systems are in place to ensure proper inflation of the cuff (typically to have a small leak just above set PIP), and location in the airway (56, 57). Cuffed tubes provide the advantage of preventing derecruitment and ensuring adequate measurement of delivered airway pressure and volume (58).

CONCLUSIONS

Since the publication of the first PALICC Consensus Conference recommendations in 2015 (59), the definition of PARDS has become entrenched in the pediatric critical care literature. Unfortunately, during this 7-year interval, there have been no definitive findings with regard to the invasive mechanical ventilation management of the patient with PARDS. The updated recommendations presented here are largely based on limited and generally weak data and/or extrapolation from meta-analyses in the adult population. The one exception is the recommendation related to use of PEEP, which should typically be maintained at or above the lower PEEP/higher FIO_2 table from the ARDS Network protocol (22). Despite three pediatric RCTs, the evidence regarding HFOV remains controversial, as stated in this document after careful analysis of all pediatric literature. We remain hopeful that the current PALICC-2 proceedings will stimulate much needed research in the management of children with PARDS.

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*The Second Pediatric Acute Lung Injury Consensus Conference (PALICC-2) group members are listed in **Appendix 1** (http://links.lww.com/PCC/C304).*

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